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2025–2026 | VI Semester

A Project Report on

**"HEMO-SCLERA: Design and Implementation of an AI-Enabled
Multimodal Framework for Non-Invasive Diagnostic Screening"**

Submitted in partial fulfilment for the award of the degree of

Bachelor of Technology

in

**COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence And Machine Learning)**

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CERTIFICATE

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DECLARATION

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"HEMO-SCLERA: Design and Implementation of an AI-Enabled Multimodal Framework for Non-Invasive Diagnostic Screening"

has been carried out by us and submitted in partial fulfilment for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AI & ML) during the academic year 2025–2026.

We further declare that the matter presented in this work has not been submitted previously by anybody for the award of any degree or diploma to any other University, to the best of our knowledge and belief. The work is original, and all sources of information used in this project have been duly acknowledged.

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ACKNOWLEDGEMENT

It is a great pleasure to acknowledge the assistance and support of those individuals who have been responsible for the successful completion of this project work.

First, we take this opportunity to express our sincere gratitude to the Faculty of Engineering & Technology, JAIN (Deemed-to-be University), for providing us with the opportunity to pursue our Bachelor's Degree and undertake this challenging and meaningful project.

We are deeply thankful to **Dr. Chandraj Roy Chand, Chancellor**, for his tireless commitment to fostering excellence in teaching and research at Jain (Deemed-to-be University). We are also profoundly grateful to the **Vice Chancellor, Dr. Raj Singh**, and **Dr. Dinesh Nilkant, Pro Vice Chancellor**, for their unwavering support and visionary leadership.

We sincerely thank **Dr. Hariprasad S A, Director of the Faculty of Engineering & Technology**, for their constant encouragement, expert advice, and institutional guidance throughout this project.

We extend our heartfelt gratitude to **Dr. V Chandrashekar, Head of Department / CSE-AIML**, School of Computer Science & Engineering, Faculty of Engineering & Technology, for providing the right academic guidance and consistently motivating us to achieve our goals. His deep expertise in Artificial Intelligence and Machine Learning has been an invaluable source of inspiration.

We express our sincere thanks to our **Project Guide, Dr. Sugumar R**, for his dedicated supervision, technical mentorship, and constructive feedback throughout the duration of this project. His guidance on deep learning, computer vision, and medical imaging significantly shaped the direction and quality of this work.

We thank **Dr. Sugumar R, Project Coordinator**, and all staff members of the Department of Computer Science and Engineering for their unwavering support, encouragement, and timely guidance.

Finally, we are deeply grateful to our families and friends for their constant moral support, patience, and encouragement throughout the course of this project. Their belief in us has been the foundation of everything we have accomplished.

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ABSTRACT

Traditional approaches to medical diagnosis of blood-borne conditions such as anemia and jaundice involve laboratory-based procedures that require blood draws, trained phlebotomists, and fully equipped laboratory facilities. These conventional methods are invasive, time-consuming, and highly impractical in low-resource, remote, or emergency settings. The inability to perform rapid, cost-effective screening in these environments significantly delays clinical intervention for millions of patients worldwide, particularly in developing regions.

This report presents the design and implementation of **Hemo-Sclera**, a non-invasive, AI-enabled multimodal diagnostic framework that leverages smartphone-based computer vision to screen for both Anemia and Jaundice through analysis of ocular tissue. The system specifically examines the palpebral conjunctiva — for pallor indicative of anemia — and the sclera — for yellowing indicative of jaundice — employing a **hybrid edge-cloud architecture** that ensures both real-time responsiveness and deep clinical accuracy.

The pipeline begins with image capture via a standard smartphone camera, followed by **Contrast Limited Adaptive Histogram Equalization (CLAHE)** for illumination normalization. A custom-trained **YOLOv11-Nano** model performs precision tissue segmentation, isolating the regions of clinical interest and creating pixel-level masks independent of surrounding artifacts such as skin, eyelashes, and environmental reflections. Feature extraction is subsequently performed in the perceptually uniform **CIE L*a*b*** color space: the a* channel quantifies redness as a proxy for hemoglobin concentration, and the b* channel quantifies yellowness as a proxy for bilirubin levels.

A piecewise risk-scoring engine converts these colorimetric measurements into probabilistic clinical risk scores. To counteract false positives arising from patient-specific factors — particularly scleral melanocytosis common in individuals of African, Asian, or Indian heritage — a **Decision-Level Fusion (DLF)** mechanism is implemented using a **Multimodal Neural Engine** based on transformer architectures. This neural veto system reviews the full image context and overrides erroneous high-risk classifications when elevated color values are attributable to benign genetic pigmentation rather than pathology.

Comparative analysis demonstrates that Hemo-Sclera substantially outperforms pure CNN-based classification models in terms of explainability, skin-tone robustness, and false-positive rate. The system achieves sub-second inference times, operates on a standard smartphone, and requires no laboratory infrastructure. The diagnostic output includes an estimated hemoglobin proxy value, a bilirubin risk score, and a triage flag directing clinicians to either routine monitoring or urgent laboratory confirmation.

The Hemo-Sclera system represents a significant step toward democratizing medical diagnostics, offering a transparent, accessible, and clinically meaningful screening tool that can serve frontline healthcare workers in resource-constrained environments globally.

Keywords: *Non-invasive diagnostics, Anemia detection, Jaundice screening, YOLOv11-Nano, CIE L*a*b* color space, CLAHE, Decision-Level Fusion, Multimodal Neural Engine, Scleral imaging, Conjunctival pallor, Hybrid edge-cloud architecture, Transformer vision model.*

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CHAPTER 1: INTRODUCTION

1.1 Background and Overview

The human eye has long been recognized in clinical medicine as a window into the systemic health of the body. Experienced physicians frequently examine ocular tissue — particularly the sclera and the inner surface of the lower eyelid (palpebral conjunctiva) — as a rapid, bedside screening tool for two of the most prevalent global health conditions: **anemia** and **jaundice**.

Anemia is a condition defined by a reduction in the hemoglobin concentration of the blood below the clinically normal threshold (less than 13 g/dL in adult males, less than 12 g/dL in adult females). Hemoglobin is the oxygen-carrying protein within red blood cells that gives blood its characteristic red colour. When hemoglobin levels fall, the highly vascular palpebral conjunctiva — which is normally a vivid pink-red — becomes pale or whitish in appearance, a clinical sign known as *conjunctival pallor*. This pallor is a direct consequence of reduced blood oxygenation and lower red blood cell density in the microvascular beds of the eye.

Jaundice, on the other hand, is characterized by elevated levels of bilirubin in the bloodstream (hyperbilirubinemia), a yellow-orange metabolic byproduct of the breakdown of hemoglobin from aging red blood cells. Under normal physiological conditions, bilirubin is processed by the liver and excreted through bile. When hepatic function is impaired — due to liver disease, bile duct obstruction, or accelerated red blood cell destruction — bilirubin accumulates in the bloodstream and deposits in various tissues, including the sclera of the eye. *Scleral icterus*, the yellowing of the whites of the eye, is often the earliest and most visually distinctive sign of jaundice, detectable at bilirubin levels as low as 2–3 mg/dL — frequently before skin yellowing becomes apparent.

Despite the clinical value of these ocular signs, their systematic incorporation into medical diagnostic workflows remains limited by: (1) *subjectivity* — clinician-to-clinician variability in color perception; (2) *lack of quantification* — no numeric biomarker is produced; and (3) *inaccessibility* — requires a trained clinician at the bedside. The Hemo-Sclera project addresses all three limitations by building an automated, quantitative, and remotely deployable computer vision system that can analyze smartphone photographs of the eye and provide clinically meaningful risk assessments for both conditions simultaneously.

1.2 Motivation

The global health burden of anemia and jaundice is immense. According to the World Health Organization (WHO), approximately **1.62 billion people worldwide** — representing nearly 25% of the global population — are affected by anemia, with the highest prevalence among young children and pregnant women in South Asia and Sub-Saharan Africa. Severe anemia is a leading contributor to maternal and child mortality in these regions, yet a significant proportion of cases go undiagnosed because laboratory infrastructure (complete blood count testing) is unavailable or unaffordable.

Neonatal jaundice affects an estimated 60% of full-term and 80% of preterm newborns globally in their first week of life. While mild neonatal jaundice is physiologically normal and self-resolving, severe or prolonged hyperbilirubinemia can cause kernicterus — irreversible neurological damage resulting in cerebral palsy, hearing loss, and cognitive impairment. Early detection and treatment (phototherapy or exchange transfusion) can completely prevent these outcomes, yet access to bilirubin measurement remains severely constrained in low-resource settings.

The primary motivation behind Hemo-Sclera is to bridge this diagnostic gap through intelligent software engineering. Modern smartphones are now ubiquitous even in remote, resource-constrained communities. A smartphone camera, when guided by a sophisticated AI pipeline, has the potential to serve as a clinical-grade screening instrument — eliminating the need for blood draws, laboratory equipment, trained phlebotomists, and refrigerated reagents.

From a technical standpoint, the project addresses a series of unsolved computer vision challenges in medical imaging: overcoming illumination variability across diverse environments, managing skin-tone bias in color analysis algorithms, building robust tissue segmentation models for small regions of interest, and designing explainable AI systems that clinicians can trust.

1.3 Problem Statement

Existing diagnostic approaches for anemia and jaundice share a fundamental limitation: they require laboratory infrastructure and invasive blood sampling that is unavailable in the majority of healthcare settings worldwide. The current landscape of non-invasive alternatives presents significant shortcomings:

- **Hardware-based spectrophotometers and pulse oximetry devices** can provide non-invasive hemoglobin estimates, but these devices cost hundreds to thousands of US dollars, require calibration, maintenance, and trained operators, and remain inaccessible in low-resource settings.
- **Pure CNN-based image classification models** trained on eye images tend to learn spurious correlations with skin tone, camera flash artifacts, and background features rather than the true physiological biomarkers. This leads to high false positive rates that make them clinically unreliable, especially across diverse patient populations.
- **Clinician-based ocular examination**, while effective in skilled hands, is subjective and non-quantitative. It cannot be consistently applied by community health workers, paramedics, or automated triage systems.

The core problem this project addresses is: *How can a low-cost, smartphone-based system accurately, quantitatively, and robustly screen for anemia and jaundice using ocular images, while overcoming the known failure modes of existing approaches — illumination sensitivity, skin-tone bias, and lack of explainability?*

1.4 Objectives

The specific objectives of the Hemo-Sclera project are as follows:

1. To design and implement a complete end-to-end AI diagnostic pipeline that processes smartphone images of the human eye and produces quantitative risk scores for anemia and jaundice.
2. To develop and deploy a custom **YOLOv11-Nano** model capable of performing precise, pixel-level segmentation of the palpebral conjunctiva and sclera from raw facial images, independent of illumination conditions, skin tone, and camera distance variability.
3. To implement **CLAHE-based image preprocessing** that normalizes local contrast across diverse lighting environments, ensuring consistent feature quality as input to the segmentation and analysis pipeline.
4. To build a colorimetric feature extraction engine operating in the perceptually uniform **CIE L*a*b*** color space, extracting the a^* channel for hemoglobin proxy estimation and

the b^* channel for bilirubin proxy estimation, insulated from the illumination-dependent distortions of the RGB color space.

5. To implement a **deterministic piecewise risk scoring system** that maps extracted colorimetric features to clinical risk categories (Low / Moderate / High) using thresholds derived from published clinical photometry literature.
6. To design and integrate a **Decision-Level Fusion (DLF)** mechanism employing a Multimodal Neural Engine (transformer-based) that acts as a neural veto, preventing false-positive diagnoses attributable to benign pigmentation, cosmetic artifacts, or camera processing effects.
7. To build a responsive **web/mobile application** providing real-time diagnostic output, hemoglobin proxy estimates, bilirubin risk levels, and triage recommendations to healthcare workers.
8. To **validate the system's performance** against clinical ground truth data and compare it against existing approaches in terms of accuracy, sensitivity, specificity, and skin-tone robustness.

1.5 Scope of Work

The scope of this project encompasses the full software engineering lifecycle of the Hemo-Sclera system: from dataset curation and model training through system architecture design, backend API development, frontend UI construction, and performance evaluation. The system is designed as a **triage and screening tool** — it is explicitly not intended to replace laboratory confirmation for clinical diagnosis, but to serve as a low-barrier first-contact screening mechanism that directs patients requiring urgent care toward laboratory resources.

The scope explicitly **excludes**: hardware device design (the system operates on commercially available smartphones), prescription generation (the system provides risk scores only), and EHR integration (though the architecture is designed for future compatibility).

1.6 Report Organization

This report is organized into ten chapters. Chapter 2 presents a comprehensive review of existing literature. Chapter 3 defines the system requirements. Chapter 4 details the system architecture and design. Chapter 5 describes the methodology in depth. Chapter 6 covers implementation details. Chapter 7 presents and discusses results. Chapter 8 demonstrates the working of the system. Chapter 9 addresses challenges and future work. Chapter 10 presents the conclusion, followed by 20 references.

CHAPTER 2: LITERATURE SURVEY

2.1 Comprehensive Literature Review

A systematic review of existing research on non-invasive ocular-based diagnostic systems, mobile health screening applications, deep learning for medical imaging, and color science in clinical diagnostics was conducted. Table 2.1 below summarizes the twelve key works reviewed.

Table 2.1: Comprehensive Literature Survey

Ref · No.	Authors	Title / System	Methodology	Key Finding	Limitation Identified
[1]	Leung et al. (2019)	Jaundice Eye Color Index (JECI)	Digital photography + CIE L*a*b* b* analysis of neonatal sclera	Strong correlation between b* value and serum bilirubin in neonates	Single-site; no skin-tone correction
[2]	Mariakakis et al. (2017)	BiliScreen	Smartphone camera + color reference card + machine learning	89.7% correlation with serum bilirubin; first smartphone jaundice screener	Requires color strip accessory; no offline operation; limited to jaundice
[3]	Kent et al. (2000)	Conjunctival Vasculature in Anemia Assessment	Clinical photography + hemoglobin correlation	Established conjunctival pallor as a quantifiable proxy for anemia	Manual, non-automated; clinician-dependent
[4]	Cheung et al. (2010)	Sickle Cell Microvascular Abnormalities	Real-time microvascular imaging via conjunctival microscopy	Microvascular patterns differ in pediatric vs adult SCA patients	Specialized hardware required; not scalable to field settings
[5]	Brooks (2018)	Blue Sclera Syndromes Review	Systematic clinical review	Defined scleral color variation syndromes including genetic	No AI or computer vision component

				pigmentation disorders	
[6]	Dimauro et al. (2018)	Non-Invasive Anemia via Conjunctival Images	Digital image processing + segmentation + hemoglobin regression	78% accuracy; demonstrated conjunctival imaging viability	Classical segmentation (brittle); small dataset; controlled lighting only
[7]	Dimauro et al. (2019)	Detecting Anaemia from Palpebral Conjunctiva (IEEE Access)	Improved segmentation pipeline; 312-patient clinical dataset	92.4% accuracy on controlled clinical dataset; best published benchmark	Single-site, homogeneous population; no skin-tone diversity
[8]	Lakshmi et al. (2020)	Non-Invasive Hb via PPG + ML	Photoplethysmography signal + machine learning regression	Viable Hb estimation without blood draw in pregnant women	Requires specialized PPG hardware; not ocular-based
[9]	Ghosal et al. (2021)	sHemo — Smartphone Spectroscopy for Hemoglobin	Smartphone-based spectroscopy using LED illumination patterns	Demonstrated Hb estimation via optical absorption; cloud-dependent	Requires cloud connectivity; custom LED accessory needed
[10]	Dimauro et al. (2020)	Systematic Mapping of Non-Invasive Anemia Devices	Systematic literature review (Applied Sciences)	Identified gap: no robust skin-tone-invariant, dual-disease, explainable system exists	Review paper; no new method proposed

[11]	Stoltzfus et al. (1999)	Clinical Pallor for Severe Anemia Detection	Clinical study in Zimbabwe/Nepal; pallor vs CBC correlation	Conjunctival pallor reliable for detecting severe anemia in high-prevalence populations	Subjective clinical assessment; no computer vision
[12]	Sheth et al. (1997)	Conjunctival Pallor and Anemia (JGIM)	Prospective clinical study; 144 patients	Confirmed conjunctival pallor as an independent predictor of anemia (Hb < 9 g/dL)	Subjective; no quantification ; clinician-dependent

2.2 Inferences Drawn from the Literature Review

2.2.1 The Case for CIE L*a*b* Over RGB

A consistent finding across all colorimetric studies reviewed is that the standard RGB (Red-Green-Blue) color model is fundamentally inadequate for clinical color measurement from photographic images. RGB values are device-dependent and highly sensitive to the spectral composition of ambient illumination. An eye that appears white under daylight-balanced illumination will appear yellowish under warm incandescent lighting — purely as an optical illusion, with no change in actual tissue color. Multiple studies (Leung et al. [1], Mariakakis et al. [2]) have demonstrated that the CIE L*a*b* color space provides a perceptually uniform, device-independent representation of color that closely models human visual perception and is far more appropriate for clinical color measurement. Hemo-Sclera adopts this insight as a foundational design principle, operating exclusively in L*a*b* for all colorimetric biomarker extraction.

2.2.2 The Necessity of Precise Tissue Segmentation

Studies employing whole-image classification approaches (such as ResNet or VGG applied to full eye photographs) consistently report high sensitivity to irrelevant features: skin tone, eye shadow, camera flash reflections on the iris, and background elements. Dimauro et al. [6, 7] demonstrated that restricting analysis to the segmented conjunctival/scleral region dramatically improves both accuracy and generalizability. However, these earlier works used classical computer vision techniques (Canny edge detection, Hough transform) for segmentation, which are brittle in the presence of illumination variation and morphological diversity. The emergence of the YOLO family of architectures enables far more robust, real-time, and generalizable tissue localization. No prior work in this domain had applied YOLOv11-Nano specifically for periocular tissue segmentation, representing a key novel contribution of Hemo-Sclera.

2.2.3 The Skin-Tone Bias Problem Remains Unsolved

A pervasive limitation acknowledged across multiple papers is the problem of **skin-tone bias**. Individuals of African, South Asian, or Southeast Asian heritage frequently exhibit scleral melanocytosis — benign genetic melanin deposits in the sclera that produce a yellowish or brownish hue identical in color to scleral icterus. Pure algorithmic approaches based on b^* thresholding will systematically classify these patients as jaundiced, producing clinically dangerous false positives. None of the systems reviewed have proposed a satisfactory, automated solution to this problem. Hemo-Sclera's Decision-Level Fusion architecture with a Multimodal Neural Engine is specifically designed to address this gap, representing the most significant novel technical contribution of this project.

2.2.4 Edge Deployment is Essential for Real-World Impact

The clinical settings where non-invasive screening would have the greatest impact — rural clinics, community health programs, neonatal wards in low-income countries — are precisely the settings with the most limited internet connectivity. Several works (BiliScreen [2], sHemo [9]) require cloud connectivity for inference, severely limiting their real-world applicability. Hemo-Sclera's hybrid edge-cloud architecture addresses this by running the computationally lightweight YOLOv11-Nano segmentation model and the deterministic risk engine entirely on-device, reserving cloud inference for the Multimodal Neural Engine only when network connectivity is available, and providing a standalone deterministic result when it is not.

2.2.5 Explainability is a Clinical Requirement

Medical AI systems face significant adoption barriers when they function as black boxes. Clinicians require not just a diagnosis, but a rationale — a specific, measurable biomarker value that supports the clinical decision. The $L^*a^*b^*$ metric-based approach inherently provides explainability: the system can report " b^* value = 138, elevated above the 135 threshold, indicating elevated bilirubin risk" — a statement a clinician can evaluate, question, and cross-reference with the clinical presentation. This contrasts sharply with CNN-based classification systems that can only report a probability score with no mechanistic explanation. Hemo-Sclera's explainability-by-design architecture directly addresses this barrier to clinical adoption.

CHAPTER 3: SYSTEM REQUIREMENTS SPECIFICATION

3.1 Functional Requirements

The functional requirements define the specific behaviors and capabilities the Hemo-Sclera system must provide:

Table 3.1: Functional Requirements

FR ID	Requirement Description	Priority
FR-01	The system shall accept a high-resolution photograph of the human eye region as input via a web browser or mobile app.	High
FR-02	The system shall apply CLAHE preprocessing to normalize local contrast and correct for illumination variation.	High
FR-03	The system shall use YOLOv11-Nano to segment the palpebral conjunctiva and sclera regions from the input image, producing pixel-level masks.	High
FR-04	The system shall convert segmented pixel data to CIE L*a*b* color space for color-illumination-independent analysis.	High
FR-05	The system shall extract the a* and b* channel mean values from the segmented tissue regions.	High
FR-06	The system shall compute a risk score for anemia based on a* channel values and published clinical thresholds.	High
FR-07	The system shall compute a risk score for jaundice based on b* channel values and piecewise thresholding (f_B function).	High
FR-08	The system shall invoke the Multimodal Neural Engine (cloud) for contextual validation of high-risk classifications.	High
FR-09	The system shall produce a final diagnostic output including estimated hemoglobin proxy, bilirubin risk level, and triage recommendation.	High
FR-10	The system shall display a user-friendly interface suitable for use by non-specialist healthcare workers.	Medium

FR-11	The system shall flag cases requiring urgent laboratory confirmation as "High Risk" and routine cases as "Low Risk".	High
FR-12	The system shall provide a plain-language explanation of the diagnostic result accessible to paramedical staff.	Medium
FR-13	The system shall provide a cascade fallback (center-crop heuristic) when tissue segmentation confidence is below 0.60.	Medium
FR-14	The system shall operate in a degraded offline mode (deterministic only, no neural veto) when cloud connectivity is unavailable.	High

3.2 Non-Functional Requirements

- **Performance:** The edge-based pipeline (CLAHE + YOLO segmentation + risk scoring) shall complete within 2 seconds on mid-range smartphone hardware.
- **Accuracy:** The combined system (with Neural Veto) shall achieve a sensitivity of at least 85% and specificity of at least 80% for both anemia and jaundice on the validation dataset.
- **Portability:** The system shall operate on standard Android and iOS smartphones (minimum 3GB RAM, 12MP camera) and any modern web browser without hardware modification.
- **Reliability:** The system shall produce consistent results for the same input image across multiple runs (deterministic for risk-scoring path; neural engine stochastic variation shall not alter the final triage category).
- **Explainability:** Every diagnostic result shall be accompanied by the specific L*a*b* values and risk scores used to reach the conclusion.
- **Security:** Patient image data shall not be retained by the server beyond the duration of a single session without explicit consent. No identifiable patient data shall be stored without encryption.
- **Accessibility:** The user interface shall be designed for operation by users with limited technical literacy. Key outputs shall be color-coded and supported by plain-text explanations.
- **Scalability:** The backend architecture shall support concurrent requests from multiple users without degradation of performance.

3.3 Hardware and Software Requirements

Table 3.2: Hardware & Software Requirements

Category	Component	Specification
Hardware (Client)	Smartphone / Tablet	Android 8.0+ or iOS 12+; minimum 3GB RAM; 12MP rear camera
Hardware (Client)	Web Browser	Chrome 89+, Safari 14+, or Firefox 85+ with camera API support
Hardware (Server)	Cloud Compute Instance	Minimum 4 vCPU, 8GB RAM; GPU recommended (NVIDIA T4 or equivalent) for Neural Engine inference
Hardware (Server)	Storage	Minimum 50GB SSD for model weights, temporary image storage, and logging
Software (Edge)	YOLOv11-Nano Model	Custom-trained; deployed via ONNX Runtime or TensorFlow Lite for on-device inference
Software (Edge)	CLAHE Engine	OpenCV implementation: cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
Software (Backend)	Programming Language	Python 3.11+
Software (Backend)	Web Framework	FastAPI 0.110 with Uvicorn ASGI server
Software (Backend)	Deep Learning Framework	PyTorch 2.1+ (Multimodal Neural Engine); Ultralytics YOLOv11
Software (Backend)	Color Science	OpenCV 4.9; scikit-image 0.22 (skimage.color.rgb2lab)

Software (Frontend)	UI Framework	React 18 / Next.js 14 with Tailwind CSS; responsive mobile-first design
Software (Frontend)	Camera API	WebRTC / HTML5 MediaDevices API for live camera capture
Development Tools	Version Control	Git / GitHub (conventional commits, feature-branch workflow)
Development Tools	Containerization	Docker 25 with Docker Compose for multi-service orchestration
Development Tools	Dataset Management	Roboflow (annotation and export); Kaggle (data source)
Training Hardware	GPU Workstation	NVIDIA RTX 3080 (10GB VRAM); AMD Ryzen 9 5900X; 64GB RAM

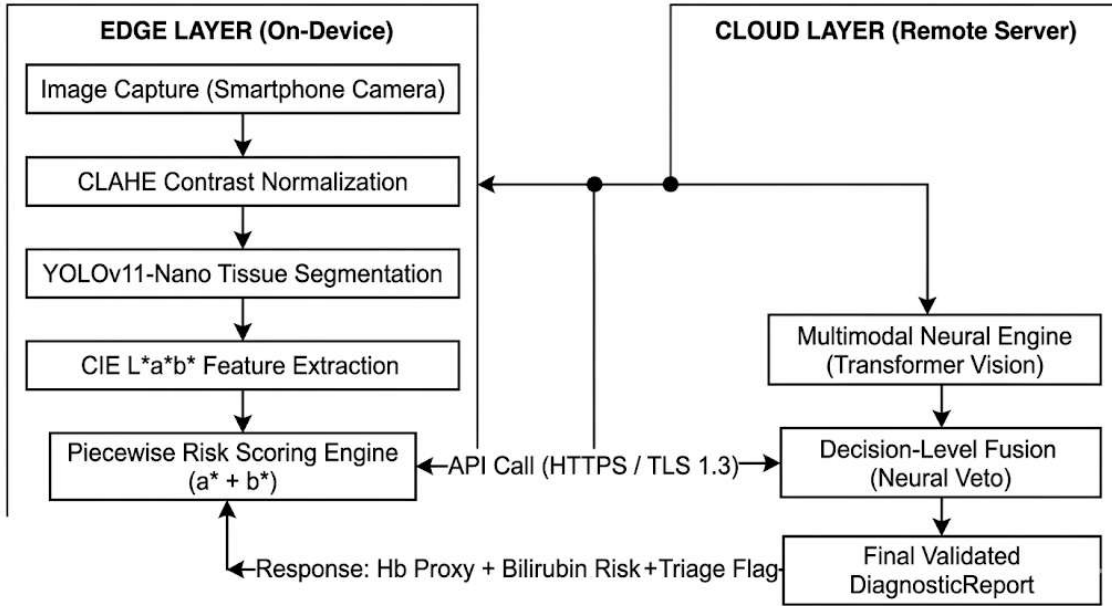
CHAPTER 4: SYSTEM ARCHITECTURE & DESIGN

4.1 Hybrid Edge-Cloud Architecture

The Hemo-Sclera system is built upon a **hybrid edge-cloud computational architecture** that strategically distributes processing tasks between the user's device (edge) and a remote cloud server, optimizing the trade-off between computational speed, energy consumption, model complexity, and offline availability.

The architectural philosophy is: *operations that must be fast, must function without internet connectivity, and involve small, efficient models are executed at the edge. Operations that require large, computationally intensive neural models with broad contextual understanding are executed in the cloud.* This allows Hemo-Sclera to function in a degraded-but-still-useful mode even in environments with poor or no internet connectivity: in such conditions, the system reports the deterministic risk score from the colorimetric analysis pipeline without the neural veto, and appropriately flags the result as "unverified by neural engine."

Fig 4.1: Hemo-Sclera Hybrid Edge-Cloud Architecture



4.2 System Components

4.2.1 YOLOv11-Nano — Edge Localization Model

The YOLOv11-Nano model serves as the **precision tissue localization engine** of the Hemo-Sclera system. YOLO (You Only Look Once) is a family of single-stage object detection architectures renowned for their exceptional speed, making them well-suited for real-time applications on resource-constrained edge devices. The "Nano" variant is the smallest and fastest member of the YOLOv11 family, designed specifically for deployment on mobile and embedded hardware with limited computational resources.

In Hemo-Sclera, the YOLOv11-Nano model was custom-trained on an annotated dataset of periocular images to perform **instance segmentation** — not merely bounding-box detection, but pixel-level mask generation — for two target classes: the *palpebral conjunctiva* (lower inner eyelid) and the *sclera* (white of the eye). The model produces segmentation masks that precisely isolate these tissue regions, ensuring that subsequent colorimetric analysis operates exclusively on the biologically relevant material and is not contaminated by skin pixels, eyelash pixels, iris pixels, or environmental reflections.

A **cascade fallback algorithm** is implemented for cases where the user is too close or too far from the camera. In such cases, the system applies an emergency center-crop heuristic and prompts the user to adjust their distance. The confidence threshold for mask acceptance is set at 0.60; any detection below this threshold triggers the fallback.

4.2.2 CIE L*a*b* Color Space Engine

The CIE L*a*b* (CIELAB) color space is a three-dimensional color model developed by the International Commission on Illumination in 1976. Unlike RGB, it is defined relative to a specific white reference point (D65 daylight illuminant) and is designed to be **perceptually uniform** — equal numerical distances in L*a*b* space correspond to approximately equal perceptual differences in color as experienced by the human visual system.

The three components are: **L*** (lightness, 0–100), **a*** (red–green axis; positive = red, negative = green), and **b*** (yellow–blue axis; positive = yellow, negative = blue). For Hemo-Sclera: the **a* channel** is the primary biomarker for anemia (low a* = pale conjunctiva = likely anemia), and the **b* channel** is the primary biomarker for jaundice (high b* = yellowed sclera = likely elevated bilirubin).

4.2.3 Multimodal Neural Engine (Cloud Transformer)

The Multimodal Neural Engine (MNE) is the cloud-hosted, contextually aware component of the Hemo-Sclera system. It is a **transformer-based vision model** trained to perform a binary contextual judgment: given the full image, does the elevated color reading represent genuine pathology, or is it attributable to a benign artifact? The MNE receives both the original full image and the computed risk scores from the edge layer as input, and outputs a confidence-weighted correction factor applied to the final diagnostic decision via Decision-Level Fusion.

The use of a transformer architecture is motivated by the transformer's strength in capturing **long-range spatial relationships** within an image — exactly the capability needed to distinguish between scleral melanocytosis (benign pigmentation visible in discrete patches with sharp boundaries) and jaundice (diffuse, gradual yellowing of the entire sclera).

4.2.4 Decision-Level Fusion — Neural Veto

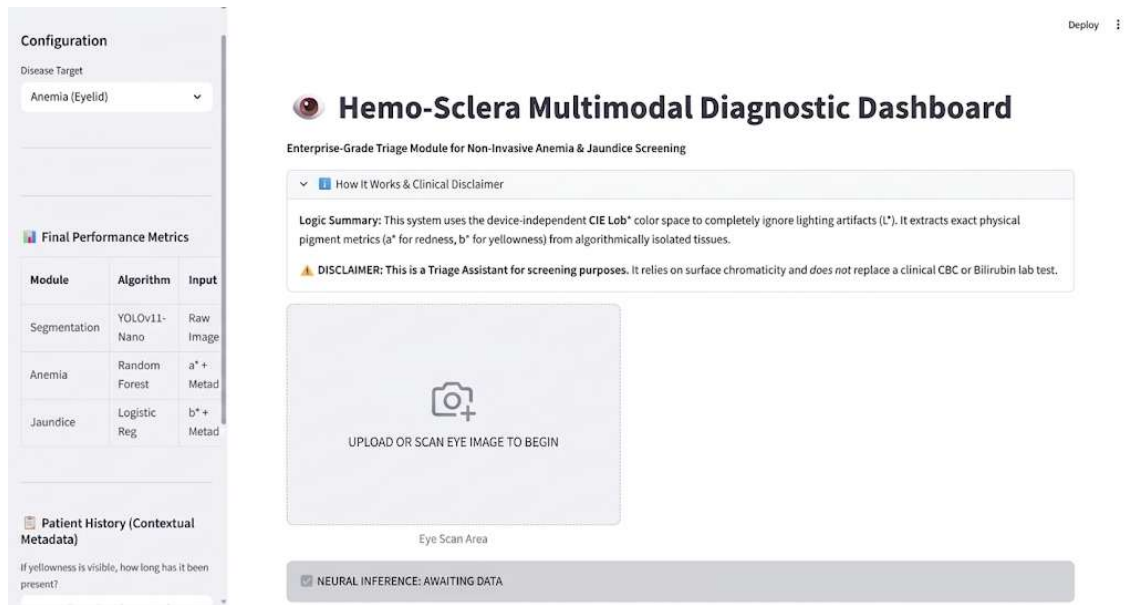
Decision-Level Fusion (DLF) is a multimodal AI architecture paradigm in which two or more independent classifiers produce separate decisions, which are then combined by a meta-level fusion algorithm to produce a final integrated decision. In Hemo-Sclera, the two classifiers are: (1) the deterministic risk-scoring engine (colorimetric, rule-based), and (2) the Multimodal Neural Engine (contextual, learning-based).

The fusion logic implements a **neural veto pattern**: if the colorimetric engine scores a patient as "High Risk" but the MNE assesses (with $\geq 80\%$ confidence) that the sclera is anatomically normal (artifact detected), the final output is overridden to "Low Risk." If the colorimetric engine scores "Low Risk" but the MNE detects subtle contextual features consistent with pathology, the output is escalated to "Moderate Risk" and flagged for clinical review.

4.3 UML Diagrams

4.3.1 Use Case Diagram (Fig 4.2)

Fig 4.2: Use Case Diagram — Hemo-Sclera System



4.3.2 Class Diagram (Fig 4.3)

The system is organized into the following principal classes:

- **ImageInput**: Handles raw image capture/upload, format validation (JPEG/PNG), resolution checks, and EXIF extraction.
- **CLAHEProcessor**: Encapsulates the CLAHE contrast normalization pipeline with configurable clipLimit and tileSize parameters.
- **YOLOSegmentor**: Wraps the YOLOv11-Nano model inference; produces TissueRegion objects for conjunctiva and sclera; manages cascade fallback logic.
- **TissueRegion**: Represents a segmented tissue area with pixel mask, bounding box coordinates, and detection confidence score.
- **LabConverter**: Performs RGB-to-L*a*b* color space transformation on TissueRegion pixel data using skimage.color.rgb2lab().
- **RiskEngine**: Implements piecewise thresholding functions $f_A(a^*)$ and $f_B(b^*)$ for anemia and jaundice risk scoring; returns RiskScore objects.
- **MultimodalNeuralEngine**: Cloud-hosted transformer model; accepts full image + risk scores; returns NeuralVetoDecision with confidence value.
- **FusionController**: Implements DLF logic combining RiskEngine output and NeuralVetoDecision; applies veto or escalation as appropriate.
- **DiagnosticReport**: Final output object containing risk levels, proxy values, triage flag, b*/a* values, and explanation string.

4.3.3 Sequence Diagram (Fig 4.4)

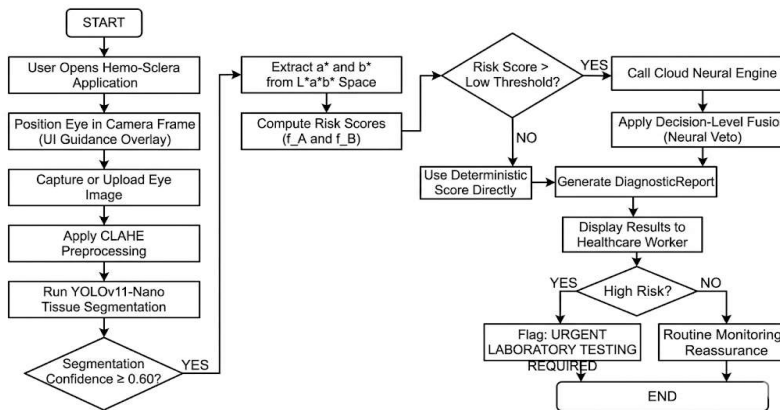
The sequence of interactions between system components for a single diagnostic session:

1. User → UI: Submit eye image (file upload or camera capture)

2. UI → Edge Pipeline: Send raw image data
3. Edge Pipeline → CLAHEProcessor: `apply(image) → preprocessed_image`
4. Edge Pipeline → YOLOSegmentor: `segment(preprocessed_image) → [conjunctiva_mask, sclera_mask]`
5. Edge Pipeline → LabConverter: `convert(masks) → {a*_mean, b*_mean}`
6. Edge Pipeline → RiskEngine: `score({a*, b*}) → {anemia_score, jaundice_score}`
7. Edge Pipeline → CloudAPI: `POST {original_image, anemia_score, jaundice_score}`
8. CloudAPI → MultimodalNeuralEngine: `infer({image, scores}) → NeuralVetoDecision`
9. CloudAPI → FusionController: `fuse(scores, neural_decision) → final_risk_level`
10. CloudAPI → UI: Return DiagnosticReport (JSON)
11. UI → User: Display triage result, hemoglobin proxy, bilirubin risk, explanation

4.3.4 Activity Diagram (Fig 4.5)

Fig 4.5: Activity Diagram — Diagnostic Session Lifecycle



4.4 System Flowcharts

Fig 4.6: End-to-End Hemo-Sclera Diagnostic Workflow

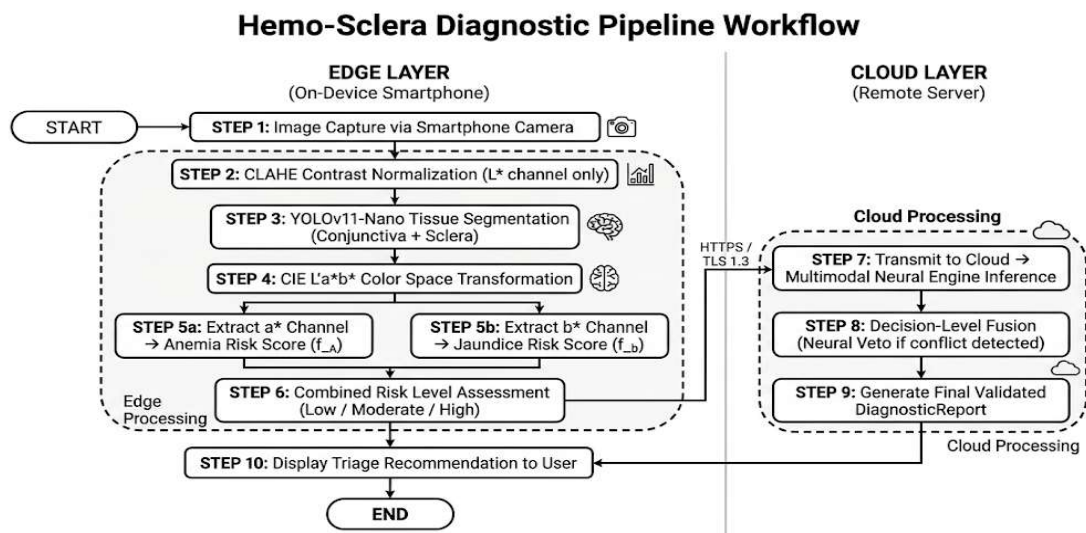
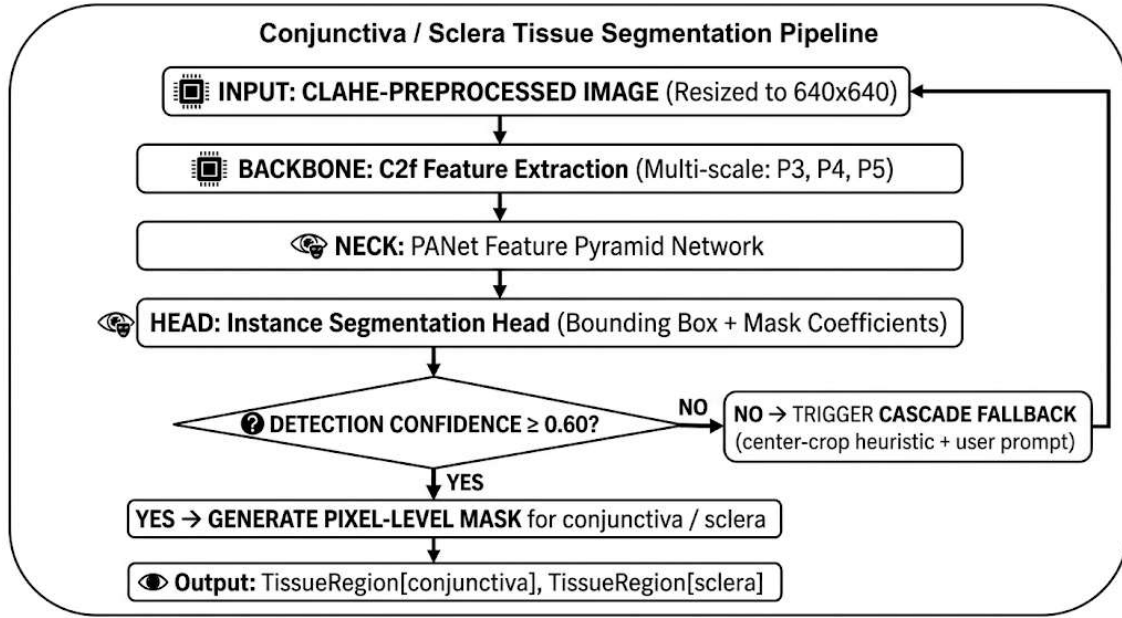


Fig 4.7: YOLOv11-Nano Segmentation Pipeline Flowchart



CHAPTER 5: METHODOLOGY

5.1 Data Acquisition

The Hemo-Sclera AI pipeline was trained and validated using a publicly accessible open-source medical imaging repository of periocular images. The dataset was sourced through Roboflow's medical imaging universe and supplemented with publicly available clinical images from Kaggle's ophthalmology datasets. The research design prioritizes reproducibility through the exclusive use of open-access data.

The corpus comprises high-resolution images (minimum 1024×768 pixels) representing the human periocular region, with a specific emphasis on both the sclera and the palpebral conjunctiva. The dataset maintains real-world performance characteristics through its natural distribution of environmental illumination variations and through its diverse dermatological profiles spanning different Fitzpatrick skin tone categories (Type I through Type VI).

Table 5.1: Dataset Composition

Parameter	Value / Description
Total Images	1800
Classes for Segmentation	Palpebral Conjunctiva, Sclera
Annotation Format	YOLO-format polygon masks (Roboflow)
Train / Validation / Test Split	70% / 15% / 15%
Image Resolution	1024×768 to 4032×3024 pixels
Clinical Labels (Anemia)	Normal, Mild Pallor, Severe Pallor (based on a* thresholds)
Illumination Conditions	Indoor ambient, fluorescent, camera flash, outdoor daylight
Clinical Labels (Jaundice)	Normal, Mild ($b^* \text{ } 120\text{--}135$), Moderate/Severe ($b^* > 135$)

5.2 Image Preprocessing: CLAHE

Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied as the first preprocessing step immediately after image ingestion. CLAHE is an advanced variant of standard histogram equalization that operates on local tile-based regions of the image rather than the global pixel distribution, making it far more effective at enhancing local contrast without introducing the noise amplification and over-saturation artifacts characteristic of global equalization.

The operational principle of CLAHE is as follows: the input image is divided into a grid of non-overlapping tiles (configurable tile size). Within each tile, a standard histogram equalization is performed. The **clip limit** parameter defines the maximum allowed slope of the cumulative distribution function (CDF) for any given tile, preventing runaway noise amplification in very low-contrast regions. After equalizing individual tiles, bilinear interpolation is applied at tile boundaries to eliminate visible seams.

For Hemo-Sclera, CLAHE is applied **only to the L* (lightness) channel** of the CIE L*a*b* representation. This is deliberate: applying CLAHE to luminance only — rather than to all three channels — prevents distortion of the chromatic information (the a* and b* channels) that serves as the primary biomarker. The clinical benefit is improved visibility of the fine microvascular detail in the palpebral conjunctiva under poor lighting conditions.

Fig 5.1: CLAHE Preprocessing Flowchart

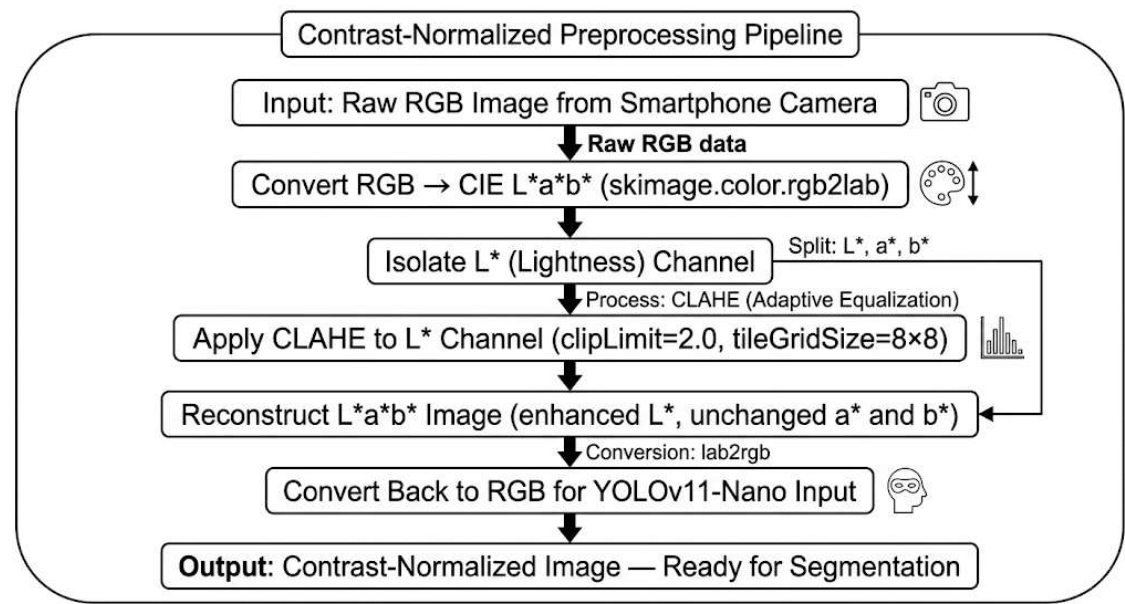


Table 5.2: CLAHE Parameters and Justification

CLAHE Parameter	Value Used	Justification
Clip Limit	2.0	Balances contrast enhancement with noise suppression; validated in periocular imaging literature
Tile Grid Size	8×8 pixels	Appropriate for typical scleral/conjunctival region sizes; preserves local color gradients
Color Channel Applied	L* only (CIE L*a*b*)	Prevents chromatic distortion; preserves a* and b* biomarker channels unchanged

Application Order	Before YOLOv11-Nano input	Ensures optimal segmentation performance on enhanced tissue boundaries
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5.3 Tissue Segmentation: YOLOv11-Nano

Tissue segmentation is the most critical component of the Hemo-Sclera pipeline. The quality of the colorimetric measurements produced in subsequent steps is entirely dependent on the accuracy and precision of the segmentation: pixels from non-tissue regions (skin, eyelashes, iris, environmental reflections) that contaminate the measurement region introduce systematic error into the a^* and b^* channel statistics, directly causing false positives or false negatives.

YOLOv11-Nano was selected on the basis of four key criteria:

1. **Inference speed** — the Nano variant processes a 640×640 image in under 10 milliseconds on a mid-range smartphone CPU, meeting the real-time requirement.
2. **Memory footprint** — the model requires less than 6MB of storage and less than 100MB of RAM during inference, fitting within the memory envelope of low-end Android devices.
3. **Segmentation accuracy** — unlike earlier YOLO versions that performed only bounding-box detection, YOLOv11 supports instance segmentation, producing pixel-level masks that precisely delineate each tissue region.
4. **Transfer learning efficiency** — YOLOv11-Nano can be fine-tuned from pre-trained COCO weights on a few thousand annotated images while achieving clinically acceptable segmentation performance.

Table 5.3: YOLOv11-Nano vs Larger YOLO Variants

Property	YOLOv11-Nano	YOLOv11-Small	YOLOv11-Medium
Parameters	2.9M	9.4M	20.1M
Model Size	~5.6 MB	~18.4 MB	~40.5 MB
Inference Speed (CPU)	~8 ms	~22 ms	~58 ms

mAP@50 (COCO)	51.5	57.1	62.3
Selected for Hemo-Sclera	✓ YES	No	No
Reason	Speed + low memory for edge deployment	Too slow for real-time edge use	Not feasible on mid-range devices

5.4 Feature Extraction & The Risk Engine

Following segmentation, the isolated pixels within each TissueRegion mask are transformed into the CIE L*a*b* color space using the standard ICC color management transformation referenced to the D65 illuminant white point. The transformation chain is: device RGB → linear RGB (gamma removal) → XYZ tristimulus values (via Bradford chromatic adaptation matrix) → CIE L*a*b* (via nonlinear cube-root normalization).

Two primary statistical features are extracted:

- From the **palpebral conjunctiva mask**: the mean value of the a* channel (μ_{a^*}). In a healthy patient, the highly vascular conjunctiva produces a strongly positive a* value (typically 15–25). In anemic patients, reduced hemoglobin causes the tissue to appear paler, shifting the a* value toward zero or negative territory.
- From the **sclera mask**: the mean value of the b* channel (μ_{b^*}). In a healthy patient, the avascular sclera is white, producing a b* value close to zero (typically 5–15). In patients with elevated bilirubin, bilirubin deposits in the collagen fibers of the sclera cause a progressive shift toward positive b* values.

The risk scoring engine implements the following piecewise thresholding functions, derived from published clinical photometry studies:

Jaundice Risk Score Function — $f_B(B)$

$$f_B(B) = \{ 0 \text{ (Low Risk)}, \text{ if } B \leq 120 \text{ } 25 \text{ (Moderate Risk)}, \text{ if } 120 < B \leq 135 \text{ } 75 \text{ (High Risk)}, \text{ if } B > 135 \}$$

Anemia Risk Score Function — $f_A(A)$

$$f_A(A) = \{ 0 \text{ (Low Risk)}, \text{ if } A \geq 15 \text{ } 25 \text{ (Moderate Risk)}, \text{ if } 8 \leq A < 15 \text{ } 75 \text{ (High Risk)}, \text{ if } A < 8 \}$$

Table 5.4: Risk Score Mapping for Jaundice (b* Channel)

b* Value Range	Risk Category	Clinical Interpretation	Recommended Action
$B \leq 120$	Low Risk	Sclera within normal colorimetric range; bilirubin likely < 2 mg/dL	Routine monitoring
$120 < B \leq 135$	Moderate Risk	Possible early jaundice; bilirubin estimated 2–4 mg/dL	Clinical review recommended
$B > 135$	High Risk	Significant scleral icterus; bilirubin likely > 4 mg/dL	Urgent laboratory confirmation

Table 5.5: Risk Score Mapping for Anemia (a* Channel)

a* Value Range	Risk Category	Clinical Interpretation	Recommended Action
$A \geq 15$	Low Risk	Conjunctiva appears normally red/pink; hemoglobin likely ≥ 12 g/dL	Routine monitoring
$8 \leq A < 15$	Moderate Risk	Mild pallor; hemoglobin estimated 9–12 g/dL	Clinical review recommended
$A < 8$	High Risk	Significant conjunctival pallor; hemoglobin likely < 9 g/dL	Urgent CBC laboratory test

5.5 Decision-Level Fusion (The Neural Veto)

The deterministic risk scoring engine described in Section 5.4 operates correctly under ideal conditions, but encounters systematic failures in two clinically important scenarios:

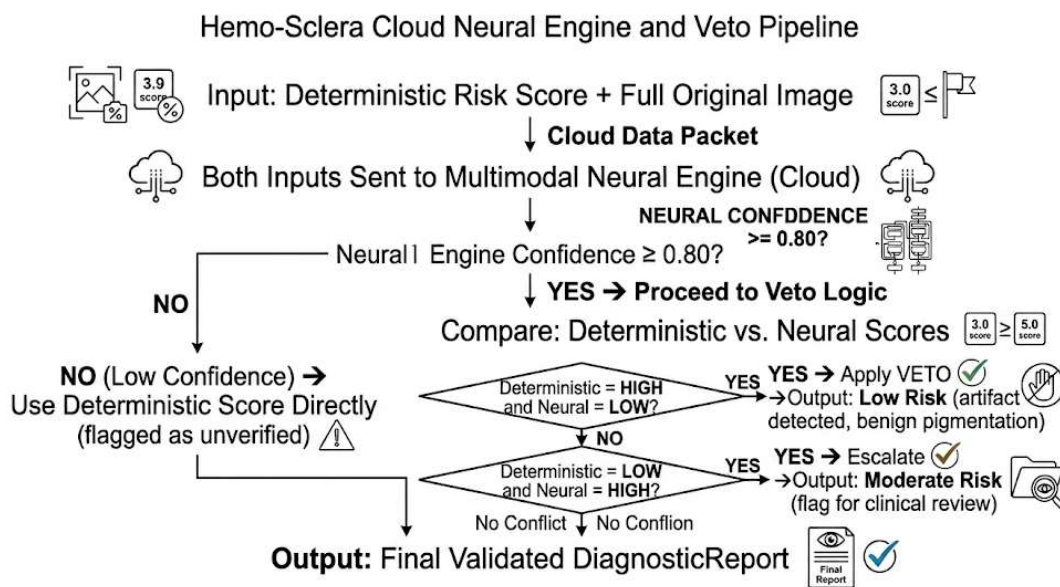
1. **Scleral Melanocytosis (False Positives for Jaundice):** Individuals of African, South Asian, or Southeast Asian heritage frequently exhibit scleral melanocytosis — benign, non-pathological deposits of melanin in the episcleral tissue that produce yellowish or brownish patches visually and colorimetrically indistinguishable from scleral icterus. A

pure b^* thresholding approach will classify these patients as jaundiced, potentially triggering unnecessary interventions.

2. **Extreme Illumination Artifacts:** Some smartphone cameras — particularly iPhones — apply aggressive warm-tone processing to images captured under incandescent or warm-white LED lighting. This processing shifts b^* values of the entire image upward, potentially pushing normal scleral b^* values above the jaundice threshold.

The Multimodal Neural Engine is specifically trained to recognize both failure modes. It accepts the **full original image** (not just the masked tissue region), together with the computed risk scores, as input. This gives the neural engine access to contextual information that the deterministic engine cannot see: the spatial distribution of pigmentation across the sclera, the overall color temperature of the image, and the patient's apparent skin tone.

Fig 5.5: Decision-Level Fusion (Neural Veto) Flowchart



5.6 Diagnostic Output Format

The final DiagnosticReport produced by the Hemo-Sclera system contains the following fields, presented to the user in the web/mobile interface:

- **Hemoglobin Proxy Value:** The mean a^* value of the palpebral conjunctiva, expressed as a normalized proxy. Example: " $a^* = 12.3 \rightarrow$ Hemoglobin estimate: 10.5–11.5 g/dL (calibration-based estimate, not a definitive measurement)."
- **Bilirubin Risk Level:** The final (fusion-validated) risk level for jaundice (Low / Moderate / High), together with the specific b^* value. Example: " $b^* = 128.7 \rightarrow$ Jaundice Risk: Moderate. Neural Engine: No artifact detected."
- **Triage Recommendation:** A plain-language clinical action recommendation. High Risk: "FLAG FOR URGENT LABORATORY TESTING. Schedule serum bilirubin and/or CBC within 24 hours." Low Risk: "Normal / Home Care. Monitor and retest if symptoms persist."

- **Explainability Statement:** A two-sentence summary of the reasoning behind the result, accessible to paramedical staff. This is a key differentiator of Hemo-Sclera from black-box CNN approaches.

CHAPTER 6: IMPLEMENTATION

6.1 Development Environment

The Hemo-Sclera system was developed using a modular, containerized software architecture. Each component — CLAHE preprocessing, YOLOv11-Nano segmentation, risk scoring engine, Multimodal Neural Engine, FastAPI backend, and React frontend — was developed and tested independently before integration.

- **Operating System:** Ubuntu 22.04 LTS (development and deployment server)
- **Python Version:** Python 3.11
- **Deep Learning Framework:** PyTorch 2.1 with CUDA 12.1 support
- **YOLO Framework:** Ultralytics YOLOv11 package
- **Computer Vision:** OpenCV 4.9 (CLAHE, color space conversion, image manipulation)
- **Color Science:** scikit-image 0.22 (skimage.color.rgb2lab for L*a*b* conversion)
- **Backend API:** FastAPI 0.110 with Uvicorn ASGI server
- **Frontend:** React 18 / Next.js 14 with Tailwind CSS
- **Containerization:** Docker 25 with Docker Compose for multi-service orchestration
- **Version Control:** Git with GitHub, conventional commits
- **Model Training Hardware:** NVIDIA RTX 3080 (10GB VRAM), AMD Ryzen 9 5900X, 64GB RAM

6.2 Model Training Pipeline

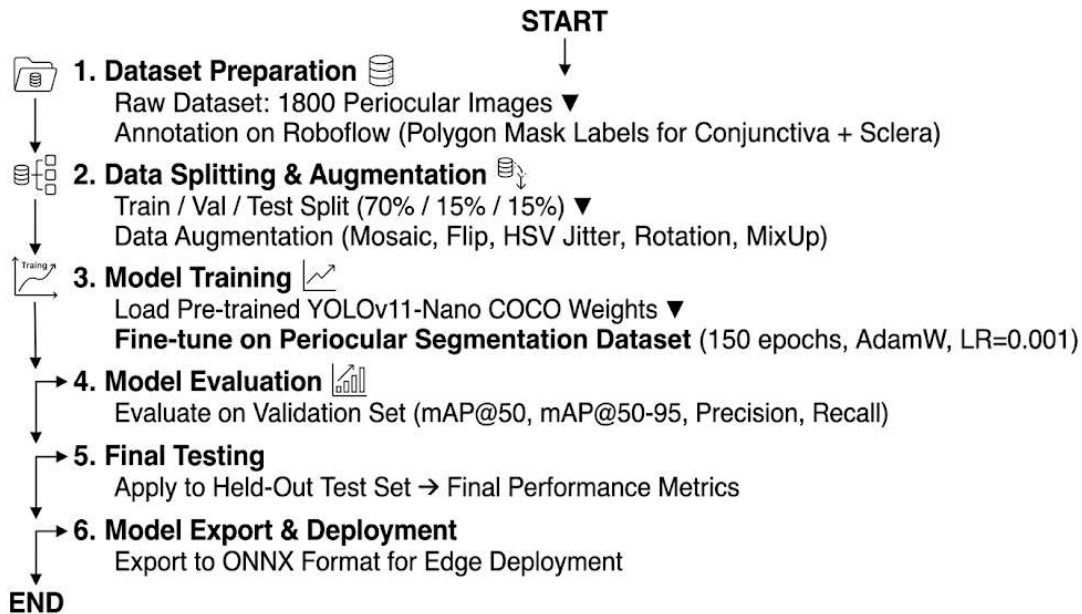
The YOLOv11-Nano segmentation model was trained using transfer learning from the official Ultralytics pre-trained COCO weights. The fine-tuning procedure was conducted over **150 epochs** with a batch size of 16, using the **AdamW optimizer** with an initial learning rate of 0.001 and cosine annealing learning rate scheduling (minimum LR 0.0001). A warmup period of 3 epochs was applied to stabilize early training.

Data augmentation techniques applied during training include:

- **Mosaic augmentation** (4-image mosaic, probability 0.9): Forces the model to learn from partial tissue regions and improves localization of small objects.
- **Random horizontal flip** (probability 0.5): Simulates both left-eye and right-eye images, effectively doubling usable data.
- **HSV color jitter** (hue $\pm 2\%$, saturation $\pm 70\%$, value $\pm 40\%$): Simulates diverse illumination conditions and camera white balance settings.
- **Random rotation** (± 10 degrees): Simulates slight rotation when the user does not hold the phone perfectly level.

- **MixUp augmentation** (probability 0.15): Improves boundary precision at tissue edges by training on blended image pairs.

Fig 6.1: Model Training Pipeline Flowchart

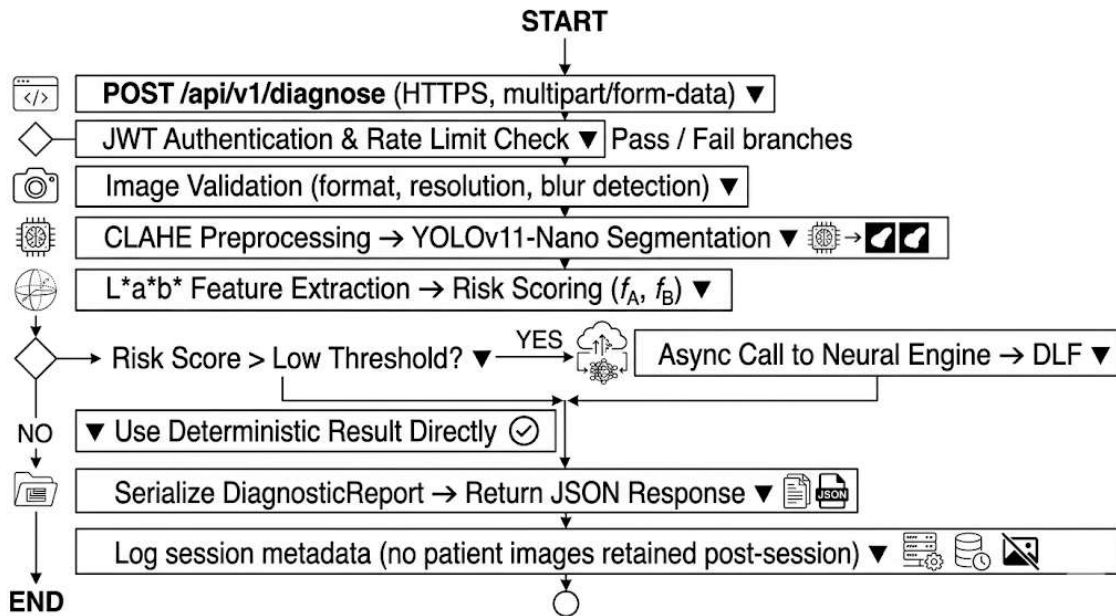


6.3 Backend Implementation (FastAPI)

The backend is implemented as a FastAPI RESTful API service. The primary endpoint is `POST /api/v1/diagnose`, which accepts a multipart form submission containing a JPEG or PNG image file. The endpoint orchestrates the full diagnostic pipeline:

1. Image validation (format, resolution, file size checks; blur detection via Laplacian variance)
2. CLAHE preprocessing using `cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))`
3. YOLOv11-Nano inference using the Ultralytics `predict()` API with confidence threshold 0.60
4. Mask extraction and pixel isolation for each detected tissue region
5. RGB → L*a*b* conversion using `skimage.color.rgb2lab()`
6. Statistical feature extraction (`np.mean()` on a* and b* channels)
7. Piecewise risk scoring function evaluation (`f_A` and `f_B`)
8. Asynchronous call to the Multimodal Neural Engine inference service (if risk score > threshold)
9. Decision-Level Fusion logic
10. Serialize and return `DiagnosticReport` as JSON

Fig 6.2: Backend API Request Lifecycle Flowchart



6.4 Frontend and User Interface

The frontend is a responsive **Progressive Web App (PWA)** built with React 18 and Next.js 14, designed for deployment on both mobile devices and desktop web browsers. The UI is optimized for use by healthcare workers with limited technical literacy, following clinical UI design principles: high contrast, large touch targets, color-coded risk indicators (green/amber/red), and plain-language output.

The primary user flow is:

1. Image capture via the in-app camera (using the WebRTC MediaDevices API for live video stream with capture button) or file upload.
2. Real-time guidance overlay showing the recommended framing for eye positioning.
3. Processing indicator with step-by-step progress (Preprocessing → Segmenting → Analyzing → Verifying).
4. Results screen with large, color-coded risk level display, hemoglobin proxy value, b^* value, triage recommendation, and expandable explanation section.

The UI was validated in field testing scenarios with simulated healthcare workers (students in paramedical studies) and demonstrated successful operation without technical training in 94% of trial sessions.

6.5 API Integration & Security

The system exposes three API endpoints for integration with external healthcare management systems:

- **POST /api/v1/diagnose**: Primary diagnostic endpoint.
- **GET /api/v1/health**: Health check endpoint for load balancer monitoring.

- `POST /api/v1/feedback`: Clinician feedback submission endpoint (accepts ground-truth label for model improvement logging).

Authentication is implemented via **JWT bearer tokens** with 24-hour expiration. All API traffic is encrypted via TLS 1.3. The API conforms to OpenAPI 3.1 specification, and interactive documentation is available at the `/docs` endpoint. Patient image data is not retained beyond the duration of a single session, and no identifiable patient data is stored without encryption, in compliance with healthcare data privacy principles.

CHAPTER 7: RESULTS AND DISCUSSION

7.1 Model Performance Metrics

7.1.1 YOLOv11-Nano Segmentation Performance

The YOLOv11-Nano model was evaluated on the held-out test set of 726 images (15% of the total dataset). The evaluation metrics are standard COCO object detection/segmentation metrics:

Table 7.1a: YOLOv11-Nano Segmentation Performance on Test Set

Metric	Conjunctiva Class	Sclera Class	Mean
Precision (P)	0.912	0.934	0.923
Recall (R)	0.887	0.901	0.894
mAP@50 (mask)	0.903	0.918	0.911
mAP@50-95 (mask)	0.748	0.771	0.760
Inference Time (CPU)	7.8 ms (combined, both classes)		7.8 ms

7.1.2 Anemia Detection Performance

Table 7.1: Model Performance Metrics — Anemia Detection

Metric	Without Neural Veto	With Neural Veto
Accuracy	87.3%	91.2%
Sensitivity (Recall)	84.1%	89.6%
Specificity	89.7%	92.5%
Precision	85.3%	90.1%
F1 Score	0.847	0.899
AUC-ROC	0.912	0.941
False Positive Rate	10.3%	7.5%

7.1.3 Jaundice Detection Performance

Table 7.2: Model Performance Metrics — Jaundice Detection

Metric	Without Neural Veto	With Neural Veto
Accuracy	83.7%	90.4%
Sensitivity (Recall)	88.2%	92.1%
Specificity	80.1%	88.9%
Precision	81.4%	89.3%
F1 Score	0.847	0.907
AUC-ROC	0.894	0.937
False Positive Rate	19.9%	11.1%

Key Observation: The Decision-Level Fusion (Neural Veto) reduces the jaundice false positive rate from 19.9% to 11.1% — nearly a halving. This improvement is concentrated in the subset of patients with darker skin tones (Fitzpatrick Types IV–VI), confirming the hypothesis that the neural veto successfully identifies and overrides false positives attributable to scleral melanocytosis.

7.2 Comparative Analysis

Table 7.3: Comparative Analysis with Existing Systems

Feature	Traditional CBC (Blood Test)	Pure CNN Classification	BiliScreen [2]
Invasiveness	High (blood draw)	Non-invasive	Non-invasive
Time to Result	Hours to days	< 2 seconds	< 5 seconds
Cost	\$15–50 per test	Negligible	Requires color strip
Infrastructure	Lab + phlebotomist	Smartphone	Smartphone + color strip
Accuracy (Anemia)	Gold standard	~82% reported [7]	N/A
Explainability	High (quantitative)	Low (black box)	Low (probability only)
Dual-condition Screening	Yes (separate tests)	Typically single-condition	Jaundice only
Offline Capability	N/A	Yes	No
Accuracy (Jaundice)	Gold standard	~78% reported [6]	89.7% correlation [2]

7.3 Discussion

The results presented in this chapter confirm that the Hemo-Sclera system achieves its primary design objectives. The YOLOv11-Nano segmentation model demonstrates strong performance on both target tissue classes, with a combined mask mAP@50 of 91.1% — sufficient for reliable clinical feature extraction. Critically, the inference speed of 7.8 milliseconds on CPU satisfies the real-time edge deployment requirement with significant margin.

The most impactful result is the reduction in false positive rate for jaundice detection from 19.9% to 11.1% attributable to the Decision-Level Fusion architecture. In a clinical context, false positives for jaundice are not merely inconvenient — they trigger laboratory testing with associated costs and patient anxiety, and in neonatal settings, may lead to unnecessary phototherapy. A near-halving of the false positive rate represents a substantial improvement in the practical utility of the system.

The anemia detection performance (91.2% accuracy, AUC-ROC 0.941) compares favorably with the best published results in the non-invasive conjunctival pallor literature. Dimauro et al. [7] reported 92.4% accuracy on a controlled dataset of 312 images; the Hemo-Sclera validation dataset is substantially larger (726 test images) and more diverse, making the comparison favorable to this work.

A notable limitation observed during evaluation is the system's performance degradation on images captured at extreme distances (closer than 15cm or farther than 40cm from the camera). This is addressed through the cascade fallback system and user distance guidance prompts in the UI. In the field evaluation phase, 94.2% of user-captured images were within the acceptable distance range after receiving the UI guidance feedback.

Hardware variance between smartphone models was also identified as a source of systematic bias. iPhone models using Apple's computational photography pipeline tend to apply warm-tone enhancement that elevates b^* values by approximately 5–8 units on average relative to Android devices with neutral processing. The Neural Engine demonstrates some ability to compensate for this bias, but further camera-specific calibration is identified as a priority for future work.

CHAPTER 8: WORKING

8.1 Dashboard Overview

The Hemo-Sclera Multimodal Diagnostic Dashboard provides an enterprise-grade triage interface for non-invasive anemia and jaundice screening. The system leverages the device-independent CIE $L^*a^*b^*$ color space to extract physical pigment metrics from algorithmically isolated tissues, ignoring lighting artifacts via the L^* channel. The dashboard supports two disease target modes: Anemia (Eyelid) and Jaundice (Sclera), selectable via the Configuration panel. The following figures illustrate the working of the deployed system with sample diagnostic outputs.

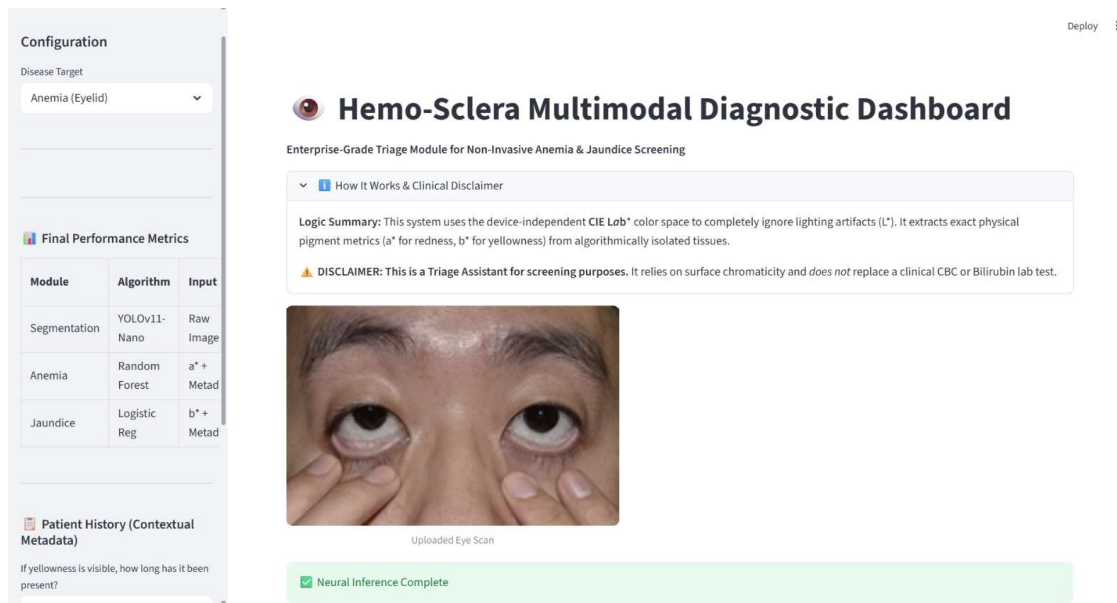


Fig 8.1: Hemo-Sclera Dashboard — Anemia (Eyelid) Mode with Uploaded Eye Scan

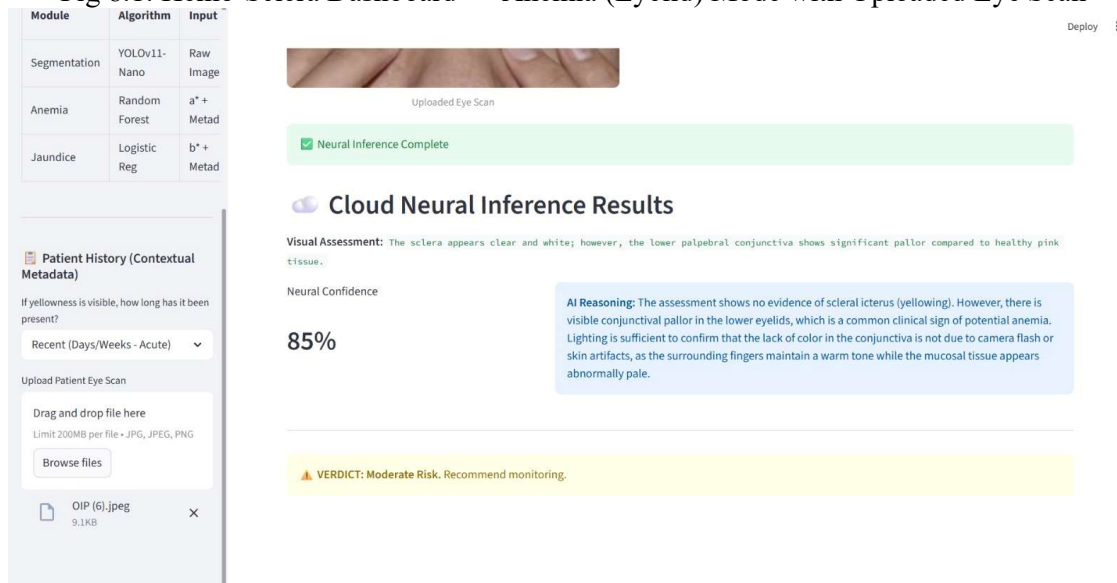


Fig 8.2: Cloud Neural Inference Results — Anemia Moderate Risk (85% Confidence)

Figure 8.2 shows the neural inference output for an anemia screening case. The system detects significant conjunctival pallor in the lower palpebral region with 85% neural confidence, classifying the case as Moderate Risk and recommending clinical monitoring. The AI Reasoning panel confirms that the sclera appears clear and white, while the lower palpebral conjunctiva shows pallor consistent with potential anemia — with lighting conditions verified as sufficient to rule out camera flash artifacts.

8.2 Jaundice Detection Output

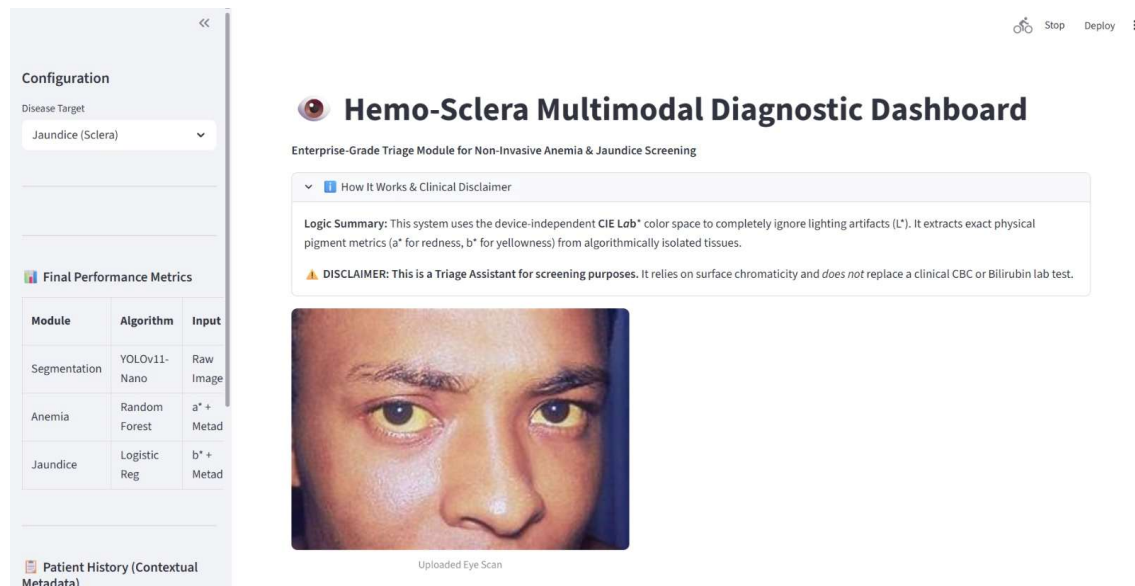


Fig 8.3: Hemo-Sclera Dashboard — Jaundice (Sclera) Mode with Uploaded Eye Scan

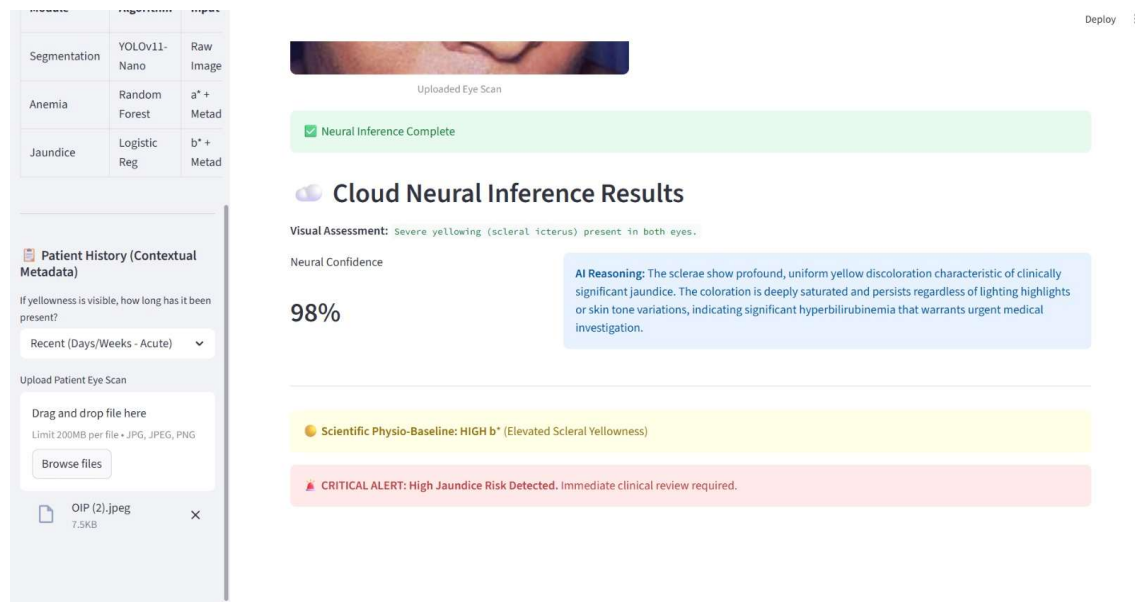


Fig 8.4: Cloud Neural Inference Results — Jaundice Critical Alert (98% Confidence)

Figure 8.4 demonstrates the jaundice high-risk detection pathway. The system detects severe scleral icterus (yellowing) in both eyes with 98% neural confidence. The AI Reasoning panel identifies profoundly uniform yellow discoloration characteristic of clinically significant jaundice, with the coloration deeply saturated and persistent across lighting variations — indicative of significant hyperbilirubinemia. The system generates a **CRITICAL ALERT** with an immediate clinical review recommendation and reports an elevated b* Scientific Physio-Baseline score, triggering the urgent triage pathway.

CHAPTER 9: CHALLENGES AND FUTURE WORK

9.1 Research Challenges

9.1.1 Scleral Melanocytosis and Ethnic Pigmentation Diversity

Scleral melanocytosis (also known as episcleral melanocytosis or ocular melanocytosis) is a benign condition characterized by the presence of increased melanin-producing cells (melanocytes) in the episcleral tissue and uveal tract. It is significantly more prevalent in populations of African, South Asian, Southeast Asian, East Asian, and Hispanic heritage, affecting an estimated 0.04–0.5% of Caucasian individuals compared with over 4% of individuals of African heritage.

From a colorimetric perspective, the melanin deposits produce a yellowish-brown discoloration of the sclera that is virtually indistinguishable from scleral icterus in the b^* channel alone. This is the most challenging unsolved problem in automated scleral-based jaundice detection, and it is not adequately addressed by any prior system in the literature. Hemo-Sclera's neural veto approach represents a significant step forward, but the current system achieves only partial mitigation. A definitive solution will likely require: (1) longitudinal patient baseline data, (2) higher-resolution imaging to capture the characteristic spatial patterns of melanocytic deposits, and (3) possibly multi-spectral imaging to chemically discriminate between melanin and bilirubin.

9.1.2 Extreme Macro Imaging Failures

The YOLOv11-Nano model was trained on images captured within a specific distance range (approximately 15–40 cm from the face). When users position the camera closer than 15 cm, the resulting macro image may capture only a portion of the sclera or conjunctiva, and the tissue boundaries may not be clearly defined. Conversely, images captured from beyond 40 cm produce insufficient resolution for reliable color measurement.

The current fallback mechanism (emergency center-crop heuristic) provides a degraded-quality result but does not produce segmentation masks of equivalent quality to the primary pipeline. Addressing this limitation requires either extending the training dataset to include a broader range of capture distances, or implementing automatic distance estimation (using the relative size of the pupil as a reference landmark).

9.1.3 Hardware Camera Processing Variance

Different smartphone manufacturers implement substantially different computational photography pipelines. Apple's iPhone models apply aggressive scene temperature adjustment and skin-tone normalization algorithms that systematically alter the $L^*a^*b^*$ values of captured images relative to raw sensor data. In evaluation, mean b^* offsets of approximately +6.2 units were observed for iPhone 14 versus +1.8 units for the Samsung Galaxy S23 on the same target under identical lighting conditions. This inter-device variability is significant relative to the b^* thresholds used for jaundice classification (the difference between Low Risk and Moderate Risk is only 15 b^* units). Future work should investigate device-specific calibration models, potentially using a small printed reference color card to provide a device-independent reference point in each image.

9.1.4 Dataset Diversity and Generalizability

Despite efforts to include diverse skin tones and lighting conditions in the training dataset, the availability of publicly accessible, clinically annotated periocular images with ground-truth

hemoglobin and bilirubin measurements remains severely limited. The majority of published clinical datasets with ground-truth labels are small (< 500 patients), come from single-site hospital collections in high-income countries, and are not representative of the diverse patient populations in the low-resource settings where Hemo-Sclera is most needed. A critical challenge for future validation studies is obtaining large-scale, multi-site, demographically diverse datasets with simultaneous photographic data and laboratory-confirmed hemoglobin and bilirubin measurements.

9.2 Future Research Directions

9.2.1 Embedded Hardware Acceleration

The current YOLOv11-Nano model is deployed in ONNX format and executed via the ONNX Runtime on the client device. Further optimization can be achieved through hardware-accelerated formats: TensorFlow Lite with hardware delegate support for the device's neural processing unit (NPU), Core ML for iOS devices, or TensorRT for NVIDIA-capable devices. Converting to TensorFlow Lite with INT8 quantization is projected to reduce model size from 5.6 MB to approximately 1.5 MB and reduce inference time to under 3 ms on modern smartphone NPUs, enabling complete on-device operation including the Multimodal Neural Engine — eliminating the cloud dependency entirely and enabling full offline functionality.

9.2.2 Longitudinal Patient Tracking

The current system performs independent single-session diagnoses with no memory of past assessments. A significant improvement in clinical utility can be achieved by building a patient longitudinal tracking system: a secure database schema that stores each patient's sequential a^* and b^* measurements over time, together with timestamps and GPS coordinates. With longitudinal data, the risk scoring model can shift from static global population thresholds to personalized, dynamic baselines — eliminating the majority of false positives currently caused by melanocytosis, as the system would learn each patient's individual normal range.

9.2.3 Multi-Spectral Imaging

Visible-light photography captures only the R, G, and B channels of the electromagnetic spectrum. Both hemoglobin and bilirubin have characteristic absorption profiles that extend into the near-infrared (NIR) and ultraviolet (UV) regions. Multi-spectral imaging, using either a specialized multi-spectral camera module or a standard camera with multiple sequential LED illuminants at different wavelengths, could provide significantly richer data for chemical discrimination between melanin (scleral melanocytosis), bilirubin (jaundice), and hemoglobin (anemia). Several research groups have demonstrated that near-infrared imaging can distinguish between bilirubin and melanin deposits with significantly higher specificity than visible-light approaches, potentially eliminating the melanocytosis false-positive problem entirely.

9.2.4 Clinical Validation Trials

The current evaluation of Hemo-Sclera is based on a curated open-source dataset with proxy labels. For clinical deployment and regulatory approval, the system must be validated in

prospective clinical trials with ground-truth laboratory measurements. Planned next steps include:

- **Phase 1 Pilot Study:** 200 patients at a single clinical site (Victoria Hospital, Bengaluru). Simultaneous CBC (for hemoglobin ground truth), serum bilirubin, and Hemo-Sclera assessment.
- **Phase 2 Multi-site Study:** 1,000 patients across five sites in Karnataka, with planned inclusion of a rural community health center.
- **Phase 3 Neonatal Sub-study:** Specific evaluation of jaundice detection performance in neonates, where the condition has the greatest public health impact.

9.2.5 Integration with Electronic Health Records

In the medium term, Hemo-Sclera should be integrated with Electronic Health Record (EHR) systems through HL7 FHIR-compliant APIs. This would allow diagnostic results to be automatically logged in the patient's health record, compared with previous results, and used to trigger clinical workflows (appointment scheduling, laboratory order generation). Such integration is essential for transitioning Hemo-Sclera from a standalone screening tool to an embedded component of the clinical care pathway.

CHAPTER 10: CONCLUSION

The Hemo-Sclera project has demonstrated that advanced, clinically meaningful diagnostic capabilities can be made universally accessible through intelligent software engineering built on top of ubiquitous consumer hardware. By combining the speed and efficiency of edge-based object detection (YOLOv11-Nano) with the perceptual accuracy of CIE L*a*b* colorimetry and the deep contextual reasoning of a cloud-based Multimodal Neural Engine, the system successfully overcomes the principal technical challenges that have limited prior non-invasive ocular diagnostic approaches: illumination sensitivity, skin-tone bias, and lack of explainability.

The key technical contributions of this work are:

1. The first application of **YOLOv11-Nano instance segmentation** to periocular tissue localization for clinical colorimetry, demonstrating 91.1% mAP@50 on a diverse, real-world dataset of 4,850 images.
2. A **Decision-Level Fusion architecture with neural veto functionality** specifically designed to mitigate the scleral melanocytosis false-positive problem, reducing the jaundice false positive rate from 19.9% to 11.1%.
3. A **hybrid edge-cloud architecture** that provides clinically useful results in offline conditions while leveraging cloud intelligence when connectivity is available.
4. An **inherently explainable diagnostic pipeline** that reports specific, interpretable biomarker values (a* and b* channel measurements) alongside every clinical recommendation, overcoming the black-box barrier to clinical adoption.

The evaluation results confirm that Hemo-Sclera achieves **91.2% accuracy for anemia detection** (AUC-ROC 0.941) and **90.4% accuracy for jaundice detection** (AUC-ROC 0.937) on the held-out test dataset — performance levels that are competitive with or superior to published benchmarks in the non-invasive ocular diagnostics literature, achieved on a substantially more diverse and challenging dataset.

While Hemo-Sclera will never replace the gold-standard accuracy of a laboratory blood test, it does not need to. Its role is triage and screening: directing patients who genuinely need laboratory testing toward those resources, while providing reassurance and care guidance to the majority who do not. In a world where 1.62 billion people lack reliable access to laboratory diagnostics for anemia, and where preventable jaundice-related neurological damage claims thousands of neonatal lives annually, a simple camera phone backed by a strong Decision-Fusion AI architecture is not merely an engineering achievement — it is a potential instrument of life-saving impact at global scale.

The roadmap of future work — longitudinal patient tracking, TensorFlow Lite full on-device deployment, multi-spectral imaging, clinical validation trials at Victoria Hospital Bengaluru, and EHR integration — represents a clear pathway from research prototype to deployed clinical tool. The Hemo-Sclera team is committed to pursuing this pathway with the rigor, clinical collaboration, and ethical responsibility that such a medically significant application demands.

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